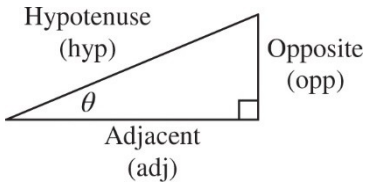
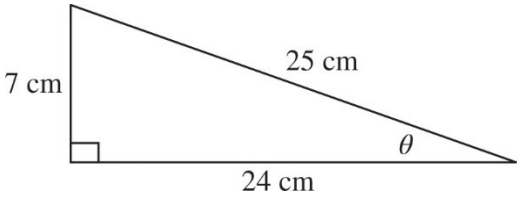
Define Trigonometric Functions of Acute Angles

In a right triangle with an acute angle θ, the longest side in the triangle is the **hypotenuse** (hyp) and is opposite the right angle.
The leg that lies on one ray of angle θ is called the **adjacent** leg (adj).
The leg that lies across the triangle from θ is called the **opposite** leg (opp).



Definition of Trigonometric Functions of Acute Angles

Function Name	Definition	Example
sine	$\sin\theta = \frac{\text{opp}}{\text{hyp}}$	$\sin\theta = \frac{7}{25}$
cosine	$\cos\theta = \frac{\text{adj}}{\text{hyp}}$	$\cos\theta = \frac{24}{25}$
tangent	$\tan\theta = \frac{\text{opp}}{\text{adj}}$	$\tan\theta = \frac{7}{24}$
cosecant	$\csc\theta = \frac{\text{hyp}}{\text{opp}}$	$\csc\theta = \frac{25}{7}$
secant	$\sec\theta = \frac{\text{hyp}}{\text{adj}}$	$\sec\theta = \frac{25}{24}$
cotangent	$\cot\theta = \frac{\text{adj}}{\text{opp}}$	$\cot\theta = \frac{24}{7}$



SOH–CAH–TOA

Sine: Opp over Hyp

Cosine: Adj over Hyp

Tangent: Opp over Adj

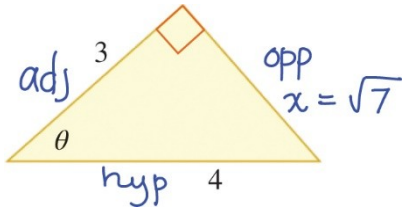
Note: The output value of a trigonometric function is unitless because the common units of length "cancel" within each ratio.

Evaluate Trigonometric Functions of Acute Angles

1. First use the Pythagorean theorem to find the length of the missing side. Then find the exact values of sin θ and cos θ.

$$x^2 + 3^2 = 4^2$$
$$x^2 = 16 - 9$$
$$x^2 = 7$$
$$x = \sqrt{7}$$

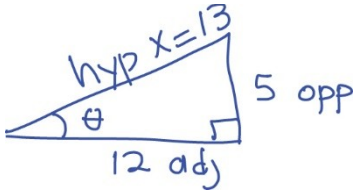
$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{\sqrt{7}}{4}$$
$$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{3}{4}$$



2. Given an acute angle θ, if $\tan\theta = \frac{5}{12}$, then find sec θ.

$$12^2 + 5^2 = x^2$$
$$144 + 25 = x^2$$
$$169 = x^2$$
$$x = 13$$

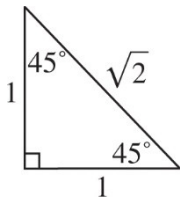
$$\sec\theta = \frac{\text{hyp}}{\text{adj}} = \frac{13}{12}$$



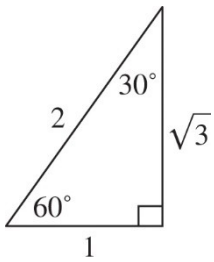
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Determine Trigonometric Function Values for Special Angles

Isosceles Right Triangle



30°–60°–90° Right Triangle



	$30° = \frac{\pi}{6}$	$45° = \frac{\pi}{4}$	$60° = \frac{\pi}{3}$
$\sin\theta$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$

1 – 2 – 3

3 – 2 – 1

step 1: write 1 2 3

step 2: $\sqrt{}$ $\sqrt{1}$ $\sqrt{2}$ $\sqrt{3}$

step 3: $\frac{}{2}$ $\frac{\sqrt{1}}{2}$ $\frac{\sqrt{2}}{2}$ $\frac{\sqrt{3}}{2}$

3. Find the exact value without the use of a calculator.

a. $\tan\frac{\pi}{3} - \sec\frac{\pi}{4}$

$= \sqrt{3} - \sqrt{2}$

b. $\csc 30° - 2\sin 45°$

$= 2 - 2\left(\frac{\sqrt{2}}{2}\right)$

$= 2 - \sqrt{2}$

Can fill from triangles

Trigonometric Function Values of Special Angles						
θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
$30° = \frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$

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$45° = \frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
$60° = \frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$

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Use Fundamental Identities

Reciprocal and Quotient Identities	
$\csc\theta = \frac{1}{\sin\theta}$ or $\sin\theta = \frac{1}{\csc\theta}$	$\sin\theta$ and $\csc\theta$ are reciprocals.
$\sec\theta = \frac{1}{\cos\theta}$ or $\cos\theta = \frac{1}{\sec\theta}$	$\cos\theta$ and $\sec\theta$ are reciprocals.
$\cot\theta = \frac{1}{\tan\theta}$ or $\tan\theta = \frac{1}{\cot\theta}$	$\tan\theta$ and $\cot\theta$ are reciprocals.
$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\tan\theta$ is the ratio of $\sin\theta$ and $\cos\theta$.
$\cot\theta = \frac{\cos\theta}{\sin\theta}$	$\cot\theta$ is the ratio of $\cos\theta$ and $\sin\theta$.

Pythagorean Identities
$\sin^2\theta + \cos^2\theta = 1$
$\tan^2\theta + 1 = \sec^2\theta$
$1 + \cot^2\theta = \csc^2\theta$

4. Given $\sin\theta = \frac{15}{17}$ and $\cos\theta = \frac{8}{17}$, use the reciprocal and quotient identities to find the value of $\tan\theta$ and $\csc\theta$.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{15}{17}}{\frac{8}{17}} \cdot \frac{17}{17} = \frac{15}{8}$$

$$\csc\theta = \frac{1}{\sin\theta} = \frac{1}{\frac{15}{17}} = \frac{17}{15}$$

5. Given $\sec\theta = \frac{\sqrt{11}}{3}$, use an appropriate Pythagorean identity to find $\tan\theta$.

$\tan^2\theta + 1 = \sec^2\theta$

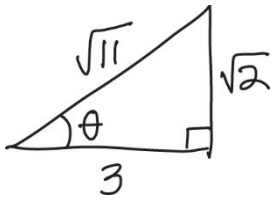
$$\tan^2\theta + 1 = \left(\frac{\sqrt{11}}{3}\right)^2$$

$$\tan^2\theta + 1 = \frac{11}{9}$$

$$\tan^2\theta = \frac{11}{9} - 1$$

$$\tan^2\theta = \frac{2}{9}$$

$$\tan\theta = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$$



Cofunction Identities		
Cofunctions of complementary angles are equal.		
$\sin\theta = \cos(90° - \theta)$ $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$	$\cos\theta = \sin(90° - \theta)$ $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$	Sine and cosine are cofunctions.
$\tan\theta = \cot(90° - \theta)$ $\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right)$	$\cos\theta = \tan(90° - \theta)$ $\cot\theta = \tan\left(\frac{\pi}{2} - \theta\right)$	Tangent and cotangent are cofunctions.
$\sec\theta = \csc(90° - \theta)$	$\csc\theta = \sec(90° - \theta)$	Secant and cosecant are

$\sec\theta = \csc\left(\frac{\pi}{2} - \theta\right)$ $\quad$ $\csc\theta = \sec\left(\frac{\pi}{2} - \theta\right)$

Examples:

$\sin 30° = \cos 60°$

$\tan 12° = \cot 78°$

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$$\csc\frac{5\pi}{12} = \sec\frac{\pi}{12}$$

6. Given $\cos 75° = \frac{\sqrt{6}-\sqrt{2}}{4}$, find a cofunction with the same value.

$$\text{complement of } 75° = 90° - 75° = 15° \rightarrow \cos 75° = \sin 15° = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Use the fundamental identities to determine if each expression is true or false. If the statement is false, provide a counterexample.

7. $\csc^2\theta \cdot \tan\theta \cdot \sin\theta \cdot \cos\theta = 1$

$$\frac{1}{\sin^2\theta} \cdot \frac{\sin\theta}{\cos\theta} \cdot \sin\theta \cdot \cos\theta = 1$$

True

8. $\tan\theta \cdot \cos\theta = 1 - \cos\theta$

$$\frac{\sin\theta}{\cos\theta} \cdot \cos\theta = \sin\theta \neq 1 - \cos\theta$$

$$\sin\frac{\pi}{6} \neq 1 - \cos\frac{\pi}{6}$$

$$\frac{1}{2} \neq 1 - \frac{\sqrt{3}}{2}$$

False

Approximate Trigonometric Functions on a Calculator

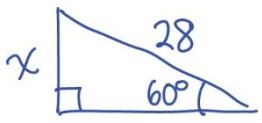
Use a calculator to approximate the function values. Round to 4 decimal places.

9. $\cos 74.82° \approx 0.2619$

10. $$\csc\frac{2\pi}{5} = \frac{1}{\sin\left(\frac{2\pi}{5}\right)} \approx 1.0515$$

Use Trigonometric Functions in Applications

11. A 28 foot slide at a water park makes an angle of 60° with the ground as it descends into a pool. What is the vertical distance from the top of the slide to the ground? Round to the nearest foot.



x = vert. distance

$$\sin 60° = \frac{\text{opp}}{\text{hyp}} = \frac{x}{28} \rightarrow$$

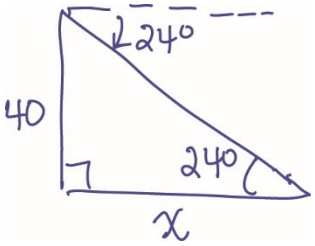
$$x = 28\sin 60°$$

$$= 28 \cdot \frac{\sqrt{3}}{2} = 14\sqrt{3} \text{ feet}$$

$$\approx 24 \text{ ft}$$

12. An observer on the roof of a 40 ft building measures the angle of depression from the roof to a park bench on the ground to be 24°. What is the distance from the base of the building to the bench as measured along the ground? Round to the nearest foot.

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x = dist to bench

$$\tan 24° = \frac{\text{opp}}{\text{adj}} = \frac{40}{x}$$

$$x = \frac{40}{\tan 24°} \approx 90 \text{ feet}$$

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